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Student Number

Knox Grammar School

2008

**Trial Higher School Certificate
Examination**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time - 2 hours
- Write using blue or black pen only
- Board approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Subject Teachers

Mr I. Bradford
Mr M. Vuletich
Mr A. Johansen
Mr J. Harnwell

Total Marks – 84

- Attempt Questions 1 – 7
- Answer each question in a separate writing booklet
- All questions are of equal value

This paper MUST NOT be removed from the examination room

Number of Students in Course: 66

Number of Writing Booklets Per Student (Four Page) 7

Total marks – 84

Attempt Questions 1-7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available

Question 1 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Find $\int_{-1}^1 \frac{2}{\sqrt{4-x^2}} dx$.

2

(b) $\int_0^{\frac{\pi}{2}} \sin^2 x \, dx$.

2

(c) The interval AB has end points $A(2, 4)$ and $B(x, y)$. The point $P(-1, 1)$ divides AB internally in the ratio 3:4. Find the coordinates of B .

2

(d) Find the size of the acute angle between the tangents to the curve $y = \tan^{-1} x$ at the points where $x = 1$ and $x = \sqrt{3}$.

Give your answer correct to the nearest minute.

(e) Solve $\frac{2}{1+2x} \geq 1$.

3

Question 2 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) A monic polynomial $P(x)$ of degree 3 has a double root at $x = 1$ and $P(2) = 13$.

2

Write $P(x)$ as a product of its factors.

- (b) Find the general solution to $2 \sin x - 1 = 0$ in terms of π .

2

- (c) Use the substitution $u = \tan x$, to evaluate $\int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{\sqrt{1 - \tan^2 x}} dx$.

3

- (d) (i) Sketch the graph of $y = 2 \cos^{-1}\left(\frac{x}{\pi}\right)$.

2

- (ii) Consider the region bounded by the curve between $x = 0$, $y = 0$ and $y = \frac{\pi}{2}$.

3

Show that $x = \pi \cos\left(\frac{y}{2}\right)$, hence find the area of this region.

Question 3 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Find the coefficient of x^{12} in the expansion of $\left(2x^2 + \frac{1}{x^2}\right)^{12}$ 2
Leave your answer in the form ${}^{12}C_r 2^k$.
- (b) (i) Express $\sqrt{3} \cos 2t - \sin 2t$ in the form $R \cos(2t + \alpha)$, where $0 < \alpha < \frac{\pi}{2}$. 2
- (ii) Hence or otherwise find all positive solutions of $\sqrt{3} \cos 2t - \sin 2t = 0$ for $0 \leq t \leq \pi$. 2
- (c) Consider the functions $f(x) = 2 \cos \frac{\pi x}{3}$ and $g(x) = 2 - x$
- (i) Sketch the graphs of $y = f(x)$ and $y = g(x)$ on the same set of axes in the domain $0 \leq x \leq 6$. 2
- (ii) Use your graph to find the number of solutions for the equation $2 \cos \frac{\pi x}{3} + x - 2 = 0$ in the domain $0 \leq x \leq 6$. 1
- (iii) Use one application of Newton's method to find a further approximation of the root near $x = 4$, for $2 \cos \frac{\pi x}{3} + x - 2 = 0$. 3
Give your answer correct to two significant figures.

Question 4 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Consider the function $f(x) = \frac{4-x^2}{1+x^2}$.
- (i) Determine the coordinates and nature of any turning points, x and y intercepts and any asymptotes. Sketch the graph of $y = f(x)$ showing these important features. **4**
 - (ii) What is the largest domain containing the value $x = 2$ for which $f(x)$ has an inverse function $f^{-1}(x)$? **1**
 - (iii) Give the equation of the inverse function, $f^{-1}(x)$ in terms of x . **2**

- (b) A particle P moves in a straight line in simple harmonic motion. The acceleration in metres per second per second is given by

$$\ddot{x} = 2 - 3x$$

where x metres, is the displacement of the particle from the origin.

Initially the particle is at $x = 1$ moving with a velocity of $\sqrt{5} \text{ ms}^{-1}$.

- (i) Using integration show that the velocity $v \text{ ms}^{-1}$ of the particle is given by **2**
$$v^2 = 4 + 4x - 3x^2.$$
- (ii) Find the amplitude of motion. **1**
- (iii) Find the centre of motion. **1**
- (iv) Find the maximum speed of the particle. **1**

Question 5 (12 marks) Use a SEPARATE writing booklet.

Marks

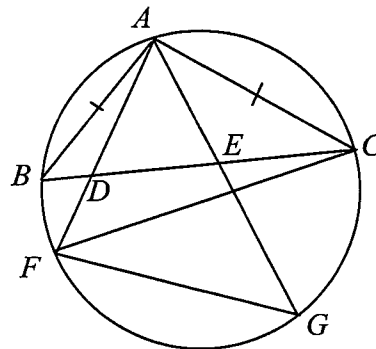
(a) The point $A(2ap, ap^2)$ lies on the parabola $x^2 = 4ay$.

(i) Show that the equation of the normal at A is $x + py = 2ap + ap^3$. 2

(ii) Given that the normal at A also passes through the point $R(-6a, 9a)$, show that $p^3 - 7p + 6 = 0$. 1

(iii) Hence, find the values of p on this parabola at which the normals to the parabola intersect at R . 2

(b)



**NOT
TO
SCALE**

The diagram shows an isosceles triangle ABC inscribed in a circle with $AB = AC$. D and E are two points on the base BC of the triangle. AD and AE are produced to meet the circle at the points F and G respectively.

(i) Copy this diagram into your writing booklet and show that $\angle ADE = \angle ACF$. 2

(ii) Show that $DEGF$ is a cyclic quadrilateral. 2

(c) Use mathematical induction to prove that for all integers $n \geq 1$ 3

$$\sum_{r=1}^n \frac{r}{2^r} = 2 - \frac{n+2}{2^n}.$$

Question 6 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) If α, β, γ are the roots of the equation $x^3 + 2x^2 - x - 5 = 0$, find the value of

2

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}.$$

- (b) The population N , of a particular species of bears in a region, after t years can be expressed as:

$$N = \frac{A}{15} + Ae^{-(\ln 2)t} \text{ where } A \text{ is a constant.}$$

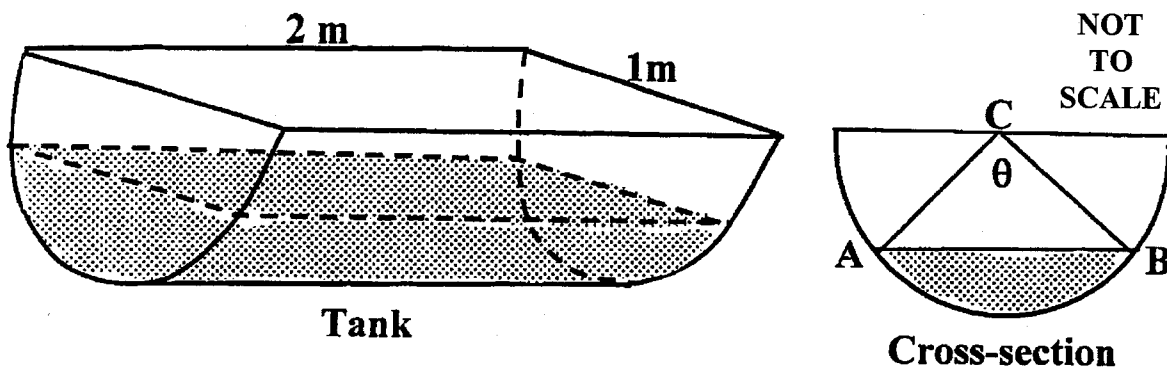
Given that the initial population was 600 bears,

- (i) find the value of A . **1**
- (ii) Find the population of bears after 10 years. **1**
- (iii) Find the time required for the population to decrease to 42 bears. **2**
- (b) The velocity $v \text{ ms}^{-1}$ of a particle at position x metres from the origin can be calculated using the equation $v = \pm \sqrt{x^3(4-x)}$.
- (i) Show that the acceleration $\ddot{x}$ is equal to $2x^2(3-x)$. **2**
- (ii) Initially the particle is 4 metres to the right of the origin.
In what direction will the particle travel immediately after leaving its initial position? **1**
- (iii) Find the maximum speed of the particle and state where it occurs. **2**
- (iv) Write a brief description of the motion of this particle as it moves from $x = 4$ to $x = 0$. **1**

Question 7 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) The diagram shows an aquarium tank 2 metres long with a semi-circular cross-section of diameter 1 metre as shown.



In the diagram of the cross-section, C is at the centre of the top edge, AB represents the water level and $\angle ACB = \theta$ where θ is measured in radians.

- (i) Show that the volume of the water in the tank is given by

1

$$V = \frac{1}{4}(\theta - \sin \theta).$$

- (ii) Show that the depth, d , of the water is given by

1

$$d = \frac{1}{2} - \frac{1}{2} \cos\left(\frac{\theta}{2}\right).$$

- (iii) Water is poured into the tank at the rate of $0.1 \text{ m}^3/\text{min}$. Find the exact rate at which the water level is rising when the depth of water is 0.2 m.

3

Question 7 Continued on page 8

Question 7 (continued)

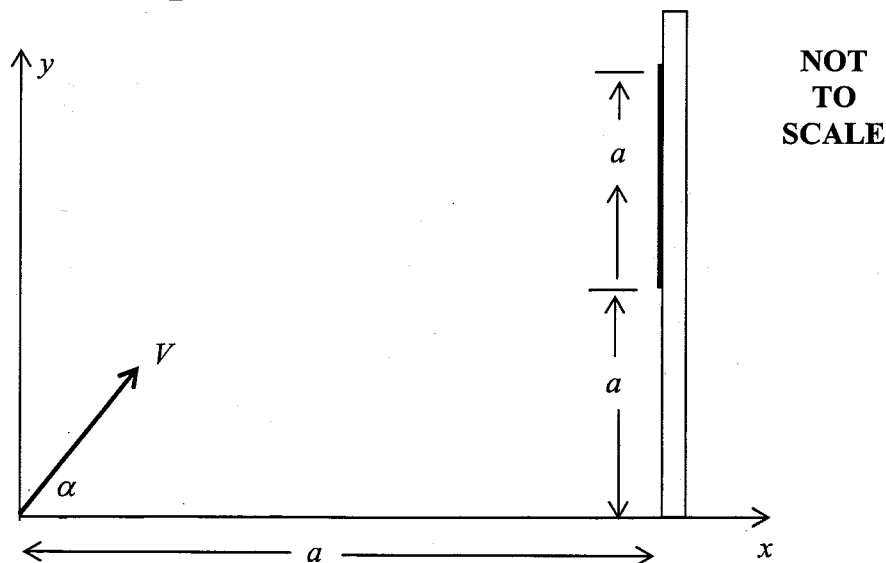
Marks

- (b) A cannon can fire a projectile with velocity $V = \sqrt{kga}$ where k and a are positive constants and at an angle α to the horizontal.

The cannon is placed on horizontal ground, a metres from a vertical building which has a large target fixed to it. The target is a metres tall with its lower edge set a metres above the ground.

Using axes as shown in the diagram, you may assume the position of the projectile, t seconds, after being fired is given by

$$x = Vt \cos \alpha, \quad y = -\frac{1}{2}gt^2 + Vt \sin \alpha, \quad \text{where } g \text{ is the acceleration due to gravity.}$$



- (i) Show that the Cartesian equation of the particle's position can be written as: 2
- $$y = x \tan \alpha - \frac{x^2}{2ka} \sec^2 \alpha.$$
- (ii) Show that the projectile will hit the base of the target if 3
- $$\tan^2 \alpha - 2k \tan \alpha + (2k + 1) = 0$$
- and hence show that if $k < 1 + \sqrt{2}$ then the projectile will always hit the building below the target.
- (iii) Given that $k = 3$, show that the target will be hit only if $3 - \sqrt{2} \leq \tan \alpha \leq 3 + \sqrt{2}$. 2

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

SOLUTIONS

KNOX EXTENSION / MATHEMATICS TRIAL 2008.

QUESTION 1

$$\begin{aligned}
 a) \int_{-1}^1 \frac{2}{\sqrt{4-x^2}} dx &= 2 \int_{-1}^1 \frac{1}{\sqrt{2^2-x^2}} dx \\
 &= 2 \left[\sin^{-1} \frac{x}{2} \right]_{-1}^1 \quad (1) \\
 &= 2 \left[\sin^{-1} \left(\frac{1}{2} \right) - \sin^{-1} \left(-\frac{1}{2} \right) \right] \\
 &= 2 \left[\frac{\pi}{6} + \frac{\pi}{6} \right] \\
 &= \frac{2\pi}{3} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 b) \int_0^{\pi/2} \sin^2 x \, dx &= \int_0^{\pi/2} \frac{1}{2} (1 - \cos 2x) \, dx \\
 &= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi/2} \quad (1) \\
 &= \frac{1}{2} \left[\frac{\pi}{2} - \frac{1}{2} \sin \pi - \left(0 - \frac{1}{2} \sin 0 \right) \right] \\
 &= \frac{\pi}{4} \quad (1)
 \end{aligned}$$

$$c) A(2,4) \quad B(x,y) \quad P(-1,1)$$

$$\frac{4(2) + 3(x)}{3+4} = -1, \quad \frac{4(4) + 3(y)}{3+4} = 1$$

$$3x + 8 = -7$$

$$x = -5$$

$$3y + 16 = 7$$

$$y = -3$$

$$\therefore B(-5, -3)$$

(1) (1)

$$d) y = \tan^{-1} x, \quad x=1, \quad x=\sqrt{5}$$

$$\frac{dy}{dx} = \frac{1}{1+x^2}$$

$$\therefore m_1 = \frac{1}{2} \quad m_2 = \frac{1}{4} \quad (1)$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan \theta = \left| \frac{\frac{1}{2} - \frac{1}{4}}{1 + \frac{1}{2} \times \frac{1}{4}} \right| \quad (1)$$

$$\tan \theta = \left| \frac{\frac{1}{4}}{\frac{9}{8}} \right|$$

$$\tan \theta = \frac{2}{9}$$

$$\theta = 12^\circ 32' \text{ (nearest min)} \quad (1)$$

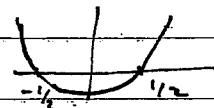
$$e) \frac{2}{1+2x} \geq 1 \quad (x \neq -\frac{1}{2})$$

$$2(1+2x) \geq (1+2x)^2$$

$$(1+2x)^2 - 2(1+2x) \leq 0 \quad (1)$$

$$(1+2x)[(1+2x)-2] \leq 0$$

$$(1+2x)(2x-1) \leq 0 \quad (1)$$



$$\therefore -\frac{1}{2} < x \leq \frac{1}{2}$$

(1)

QUESTION 2

(a) $P(x)$, monic, degree 3
double root $x=1$.

$$\therefore P(x) = (x-1)^2(x-a) \quad (1)$$

$$P(2) = 2-a$$

$$\therefore 2-a = 13$$

$$a = -11$$

$$\therefore P(x) = (x-1)^2(x+11) \quad (1)$$

(b) $2 \sin x - 1 = 0$

$$\sin x = \frac{1}{2} \quad (1)$$

$$\therefore x = n\pi + (-1)^n \sin^{-1}\left(\frac{1}{2}\right)$$

$$x = n\pi + (-1)^n \frac{\pi}{6} \quad (1)$$

(c) $\int_0^{\pi/4} \frac{\sec^2 x}{\sqrt{1-\tan^2 x}} dx$

$$u = \tan x$$

$$du = \sec^2 x dx$$

$$\text{For } x=0, u=0$$

$$x = \frac{\pi}{4}, u=1$$

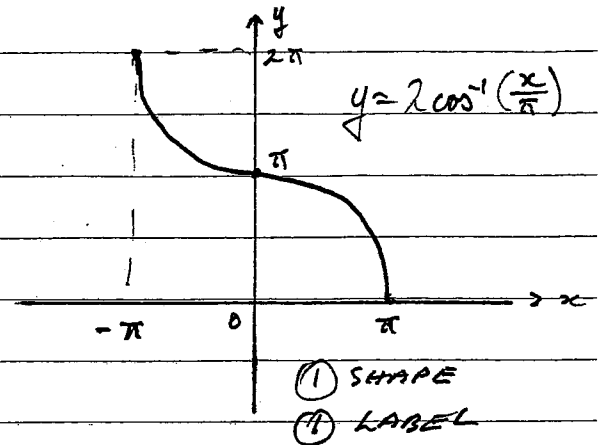
$$\therefore \int_0^{\pi/4} \frac{\sec^2 x}{\sqrt{1-\tan^2 x}} dx = \int_0^1 \frac{du}{\sqrt{1-u^2}} \quad (2)$$

$$= \left[\sin^{-1} u \right]_0^1$$

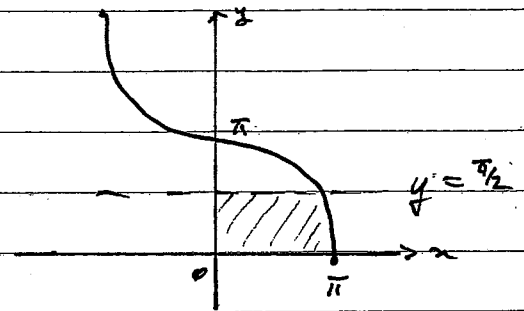
$$= \sin^{-1}(1) - \sin^{-1}(0)$$

$$= \frac{\pi}{2} \quad (1)$$

d) i)



ii)



$$y = 2 \cos^{-1}\left(\frac{x}{\pi}\right)$$

$$\frac{y}{2} = \cos^{-1}\left(\frac{x}{\pi}\right)$$

$$\frac{x}{\pi} = \cos\left(\frac{y}{2}\right)$$

$$\therefore x = \pi \cos\left(\frac{y}{2}\right) \quad (1)$$

$$\text{Area} = \int_0^{\pi/2} x dy$$

$$= \pi \int_0^{\pi/2} \cos \frac{y}{2} dy$$

$$= 2\pi \left[\sin \frac{y}{2} \right]_0^{\pi/2} \quad (1)$$

$$= 2\pi \left[\sin \frac{\pi}{4} - \sin 0 \right]$$

$$= 2\pi \times \frac{1}{\sqrt{2}}$$

$$= \sqrt{2} \pi \text{ units}^2 \quad (1)$$

QUESTION 3

(a) $(2x^2 + \frac{1}{x^2})^{12}$

$$= \sum_{k=0}^{12} {}^{12}C_k (2x^2)^k \left(\frac{1}{x^2}\right)^{12-k}$$

$$= \sum_{k=0}^{12} {}^{12}C_k \cdot 2^k \cdot x^{2k} \cdot x^{-24+2k}$$

$$= \sum_{k=0}^{12} {}^{12}C_k \cdot 2^k \cdot x^{4k-24} \quad (1)$$

$$\therefore 4k-24 = 12$$

$$k = 9$$

Coefficient of x^{12} is ${}^{12}C_9 \times 2^9 \quad (1)$

(b) i) $\sqrt{3} \cos 2t - \sin 2t$

$$R = \sqrt{(\sqrt{3})^2 + 1^2}$$

$$R = 2$$

$$\tan \alpha = \frac{1}{\sqrt{3}}, \alpha = \frac{\pi}{6}$$

$$\therefore 2 \left(\frac{\sqrt{3}}{2} \cos 2t - \frac{1}{2} \sin 2t \right)$$

$$= 2 \cos \left(2t + \frac{\pi}{6} \right) \quad (2)$$

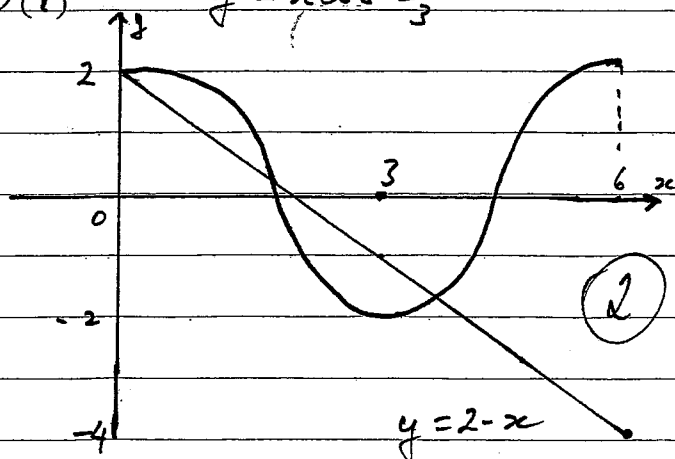
ii) $2 \cos \left(2t + \frac{\pi}{6} \right) = 0$

$$2t + \frac{\pi}{6} = \frac{\pi}{2}, \frac{3\pi}{2}, \dots \quad (1)$$

$$2t = \frac{\pi}{3}, \frac{4\pi}{3}, \dots$$

For $0 \leq t \leq \pi$, $t = \frac{\pi}{6}$ or $\frac{2\pi}{3} \quad (1)$

(c) (i) $y = 2 \cos \frac{\pi x}{3}$



ii) Number of solutions

$$2 \cos \frac{\pi x}{3} + x - 2 = 0$$

is 3 solutions. 1

iii)

$$\text{Let } h(x) = 2 \cos \frac{\pi x}{3} + x - 2$$

$$\text{If } x_1 = 4$$

$$x_2 = 4 - \frac{h(4)}{h'(4)} \quad (1)$$

$$h(4) = 2 \cos \frac{4\pi}{3} + 2$$

$$h'(x) = -\frac{2\pi}{3} \sin \frac{\pi x}{3} + 1$$

$$h'(4) = -\frac{2\pi}{3} \sin \frac{4\pi}{3} + 1$$

$$\therefore x_2 = 4 - \frac{2 \cos \frac{4\pi}{3} + 2}{-\frac{2\pi}{3} \sin \frac{4\pi}{3} + 1}$$

$$x_2 = 3.6 \quad (2 \text{ s.f.})$$

2

QUESTION 4

(a) $f(x) = \frac{4-x^2}{1+x^2}$

i) $f'(x) = \frac{-2x(1+x^2) - 2x(4-x^2)}{(1+x^2)^2}$
 $= \frac{-2x - 2x^3 - 8x + 2x^3}{(1+x^2)^2}$
 $= \frac{-10x}{(1+x^2)^2}$

$f'(x) = 0$ when $x = 0$

$f'(0-\epsilon) > 0$

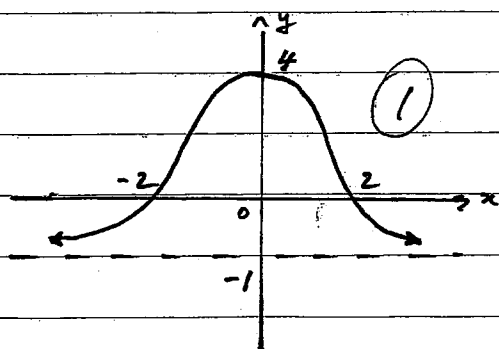
$f'(0+\epsilon) < 0$

$\therefore (0, 4)$ is a rel. maximum (2)

when $x = 0 \Rightarrow y = 4$

when $y = 0 \Rightarrow 4 - x^2 = 0$
 $x = \pm 2$

Horizontal asymptote
 at $y = -1$. (1)



ii) $x \geq 0$ (1)

iii) $y = \frac{4-x^2}{1+x^2}$

$\therefore x = \frac{4-y^2}{1+y^2}$

$x + xy^2 = 4 - y^2$

$xy^2 + y^2 = 4 - x$

$y^2(x+1) = 4-x$

$y^2 = \frac{4-x}{x+1}$

$\therefore y = \sqrt{\frac{4-x}{x+1}}$ $y \geq 0$ for inverse.

$\therefore f^{-1}(x) = \sqrt{\frac{4-x}{x+1}}$ (2)

b)

$\ddot{x} = 2 - 3x$

i) $\frac{d}{dx}(\frac{1}{2}v^2) = 2 - 3x$
 $\frac{1}{2}v^2 = 2x - \frac{3x^2}{2} + C$

$v^2 = 4x - 3x^2 + K$

when $x = 1$, $v = \sqrt{5}$

$\therefore 5 = 4(1) - 3(1) + K$

$K = 4$

$\therefore v^2 = 4x - 3x^2 + 4$

$v^2 = 4 + 4x - 3x^2$ (2)

ii) when $v = 0$, $3x^2 - 4x - 4 = 0$

$(3x+2)(x-2) = 0$

$x = -\frac{2}{3}$ or $x = 2$

Amplitude $= 2 - (-\frac{2}{3}) = \frac{4}{3} \text{ m}$. (1)

iii) Centre of motion $x = \frac{2}{3}$ (1)

iv) Max. speed when $x = \frac{2}{3}$
 Max. speed $= \frac{4\sqrt{3}}{3} \text{ m/s}$

QUESTION 5

(a) i) $x^2 = 4ay$

$$y = \frac{x^2}{4a}$$

~~$$\frac{dy}{dx} = \frac{2x}{4a}$$~~

at $x = 2ap \Rightarrow \frac{dy}{dx} = p$

Gradient normal = $-\frac{1}{p}$.

Eye. normal,

$$y - ap^2 = -\frac{1}{p}(x - 2ap) \quad (2)$$

$$py - ap^3 = -x + 2ap$$

$$x + py = 2ap + ap^3$$

ii) $(-6a, 9a)$

$$\therefore -6a + 9ap = 2ap + ap^3$$

$$ap^3 - 7ap + 6a = 0$$

$$p^3 - 7p + 6 = 0 \quad a \neq 0. \quad (1)$$

(ii) $p^3 - 7p + 6 = 0$

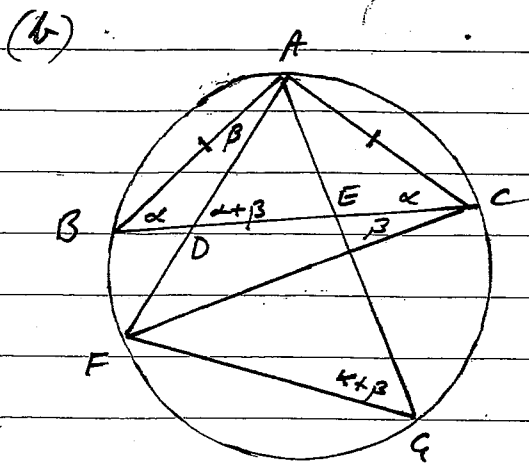
Let $P(p) = p^3 - 7p + 6$

$$P(1) = 0 \Rightarrow p=1 \text{ is a root.}$$

$$P(p) = (p-1)(p^2+p-6)$$
$$= (p-1)(p+3)(p-2)$$

$$= (p-1)(p+3)(p-2)$$

$\therefore$ Values of P are 1, 2 or -3. (2)



i) To prove $\angle ADE = \angle ACF$

Let $\angle ABD = \alpha$, $\angle BAF = \beta$

$\angle ACF = \alpha$, given $\triangle ABC$ is isosceles.

$\angle BCF = \beta$, angles in same segment on arc BF

$$\therefore \angle ACF = \alpha + \beta$$

$$\angle ADE = \alpha + \beta \text{ exterior angle } \triangle BA$$

$$\therefore \angle ADE = \angle ACF.$$

ii) $\angle ACF = \alpha + \beta$

$\angle AGF = \alpha + \beta$ angles in same

$\therefore \angle ADE = \angle AGF$ segment, over AF.

Since the exterior angle of

Qued. $DECF$ is equal to interior (2)

opposite then $DEGF$ is a cyclic quad

c) To prove $\sum_{r=1}^n \frac{r}{2^r} = 2 - \frac{n+2}{2^n}$, $n \geq 1$

For $n=1$

$$\begin{aligned} \text{LHS} &= \sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2^1} & \text{RHS} &= 2 - \frac{1+2}{2^1} \\ &= \frac{1}{2} & &= \frac{1}{2} \end{aligned}$$

Assume $\sum_{r=1}^n \frac{r}{2^r} = 2 - \frac{n+2}{2^n}$ where $n=k$.

$$\therefore \sum_{r=1}^k \frac{r}{2^r} = 2 - \frac{k+2}{2^k}$$

When $n = k + 1$

$$\sum_{r=1}^{K+1} \frac{r}{2^r} = \sum_{r=1}^K \frac{r}{2^r} + \frac{K+1}{2^{K+1}}$$

$$= 2 - \frac{k+2}{2^k} + \frac{k+1}{2^{k+1}}$$

$$= 2 - \left[\frac{2k+4 - (k+1)}{2^{k+1}} \right]$$

$$= 2 - \left[\frac{k+3}{2^{k+1}} \right]$$

$$= 2 - \frac{(k+1)+2}{2^{k+1}}$$

= RHS. when $n = k+1$
etc.

QUESTION 6.

$$(a) \quad x^3 + 2x^2 - 2x - 5 = 0$$

$$\alpha + \beta + \gamma = -2$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = -1$$

$$\alpha\beta\gamma = 5$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \alpha\gamma}{\alpha\beta\gamma}$$

$$= -\frac{1}{5} \quad (2)$$

$$(b) \quad N = \frac{A}{15} + Ae^{-(\ln 2)t}$$

$$i) \quad N = 600, t = 0$$

$$600 = \frac{A}{15} + Ae^0$$

$$600 = \frac{16A}{15}$$

$$A = \frac{1125}{2} = 562.5 \quad (1)$$

$$ii) \quad \text{when } t = 10$$

$$N = \frac{A}{15} + Ae^{-(\ln 2) \times 10}$$

$$N = 38 \quad (\text{whole number}) \quad (1)$$

$$iii) \quad \text{when } N = 42$$

$$42 = \frac{A}{15} + Ae^{-(\ln 2)t}$$

$$\frac{42 - \frac{A}{15}}{A} = e^{-\ln 2 t}$$

$$\ln \left[\frac{1}{A} \left(42 - \frac{A}{15} \right) \right] = -\ln 2 t$$

$$t = -\frac{\ln \left[\frac{1}{A} \left(42 - \frac{A}{15} \right) \right]}{\ln 2}$$

$$t = 6.966$$

$$t \approx 7 \text{ years.} \quad (2)$$

$$(b) \quad v = \pm \sqrt{x^3(4-x)}$$

$$(i) \quad v^2 = x^3(4-x)$$

$$\frac{1}{2}v^2 = \frac{1}{2}x^3(4-x)$$

$$\frac{d}{dx} \left(\frac{1}{2}v^2 \right) = \frac{1}{2} [3x^2(4-x) + (-1)x^3]$$

$$= \frac{1}{2} [12x^2 - 3x^3 - x^3]$$

$$= \frac{1}{2} [12x^2 - 4x^3]$$

$$= \frac{1}{2} \cdot 4 [3x^2 - x^3]$$

$$= 2x^2(3-x)$$

$$\therefore \ddot{x} = 2x^2(3-x) \quad (2)$$

$$ii) \quad t = 0, x = 4 \Rightarrow v = 0$$

$$\ddot{x} = -32$$

$\therefore$ travels to the left after leaving from initial position. (1)

$$iii) \quad \text{Max. speed when } \ddot{x} = 0$$

$$\ddot{x} = 0 \text{ at } x = 0 \text{ and } x = 3.$$

$$\text{at } x = 0 \Rightarrow v = 0$$

$$\text{at } x = 3 \Rightarrow v = -\sqrt{27}$$

$$\therefore \text{Max speed is } \sqrt{27} \text{ m/s at } x = 3. \quad (2)$$

(iv) The particle starts 4m to the right of the origin and accelerates left, reaching its max speed at $x = 3$ before slowing down until it reaches $x = 0$ where it momentarily stops and then accelerates to the right. (1)

Question 7.

a) i) Area of minor

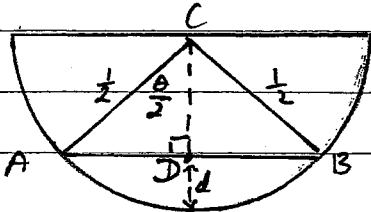
$$\text{segment AB} = \frac{1}{2} \left(\frac{1}{2}\right)^2 (\theta - \sin \theta)$$

$$= \frac{1}{8} (\theta - \sin \theta)$$

$$\therefore \text{Volume} = 2 \times \text{Area}$$

$$= \frac{1}{4} (\theta - \sin \theta) \quad (1)$$

ii)



$$CD = \frac{1}{2} \cos \frac{\theta}{2}$$

$$\therefore d = \frac{1}{2} - \frac{1}{2} \cos \frac{\theta}{2} \quad (1)$$

iii) $\frac{dv}{dt} = 0.1$, $\frac{dd}{dt} = ?$

$$\frac{dd}{dt} = \frac{dd}{d\theta} \cdot \frac{d\theta}{dt}$$

$$\frac{d\theta}{dt} = \frac{d\theta}{dv} \cdot \frac{dv}{dt}$$

$$= \frac{4}{1 - \cos \theta} \times 0.1$$

$$= \frac{2}{5(1 - \cos \theta)} \quad (1)$$

$$\therefore \frac{dd}{dt} = \frac{1}{4} \sin \frac{\theta}{2} \times \frac{2}{5(1 - \cos \theta)}$$

$$= \frac{\sin \frac{\theta}{2}}{10(1 - \cos \theta)}$$

$$\text{when } d = 0.2,$$

$$0.2 = \frac{1}{2} - \frac{1}{2} \cos \frac{\theta}{2}$$

$$0.4 = 1 - \cos \frac{\theta}{2}$$

$$\cos \frac{\theta}{2} = 0.6$$

$$\theta = 2 \cos^{-1}(0.6) \quad (1)$$

$$\therefore \frac{dd}{dt} = \frac{\sin \frac{\theta}{2}}{10(1 - \cos \theta)} \text{ when } \theta = 2 \cos^{-1}(0.6)$$

$$= \frac{\sin[\cos^{-1}(0.6)]}{10[1 - \cos(2 \cos^{-1}(0.6))]}$$

$$= \frac{\sin[\cos^{-1}(0.6)]}{10[1 - (2 \cos^2(\cos^{-1}(0.6)) - 1)]}$$

$$= \frac{0.8}{10(1 - (0.72 - 1))}$$

$$= \frac{1}{16} \text{ m/min.} \quad (1)$$

(b)

i) $x = vt \cos \alpha$, $y = -\frac{1}{2}gt^2 + vt \sin \alpha$

$$v = \sqrt{kg a}$$

$$t = \frac{x}{v \cos \alpha}$$

$$\therefore y = -\frac{1}{2}g \left(\frac{x}{v \cos \alpha}\right)^2 + \frac{vx}{v \cos \alpha} \sin \alpha$$

$$= -\frac{1}{2}g \frac{x^2}{v^2 \cos^2 \alpha} + x \tan \alpha$$

$$= x \tan \alpha - \frac{1}{2} \cdot \frac{gx^2}{kg a} \cdot \sec^2 \alpha$$

$$= x \tan \alpha - \frac{x^2}{2ka} \sec^2 \alpha. \quad (2)$$

ii) To hit base of target, $x=a$, $y=0$

$$\therefore a = a \tan \alpha - \frac{a}{2k} \sec^2 \alpha$$

$$1 = \tan \alpha - \frac{1}{2k} \sec^2 \alpha$$

$$\frac{1}{2k} \sec^2 \alpha - \tan \alpha + 1 = 0$$

$$\sec^2 \alpha - 2k \tan \alpha + 2k = 0$$

$$(1 + \tan^2 \alpha) - 2k \tan \alpha + 2k = 0$$

$$\tan^2 \alpha - 2k \tan \alpha + (2k+1) = 0 \quad (2)$$

QUESTION 7 cont'd.

$$\tan^2 \alpha - 2k \tan \alpha + 2k + 1 = 0$$

$$\Delta = 4K^2 - 4 \cdot 1 \cdot (2k+1)$$

$$= 4k^2 - 8k - 4$$

$$= 4(k^2 - 2k - 1)$$

Below target, $\Delta < 0$

$$k^2 - 2k - 1 < 0$$

$$K = \frac{2 \pm \sqrt{8}}{2}$$

$$K = 1 \pm \sqrt{2}$$

$\therefore$ below target if

(1)

$$1 - \sqrt{2} < K < 1 + \sqrt{2}$$

iii) $K=3$, $x=a$, $a \leq y \leq 2a$

$$\therefore a \leq a \tan \alpha - \frac{a}{2K} \sec^2 \alpha \leq 2a$$

$$1 \leq \tan \alpha - \frac{1}{2K} \sec^2 \alpha \leq 2 \quad (a \neq 0)$$

$$1 \leq \tan \alpha - \frac{1}{6}(1 + \tan^2 \alpha) \leq 2 \quad (K=3)$$

$$6 \leq 6 \tan \alpha - 1 - \tan^2 \alpha \leq 12$$

$$-12 \leq \tan^2 \alpha - 6 \tan \alpha + 1 \leq -6 \quad (1)$$

$$\therefore \tan^2 \alpha - 6 \tan \alpha + 13 \geq 0 \quad \text{or} \quad \tan^2 \alpha - 6 \tan \alpha + 7 \leq 0$$

No solution

$$\tan \alpha = \frac{6 \pm \sqrt{36 - 4 \cdot 1 \cdot 7}}{2}$$

$$\tan \alpha = \frac{6 \pm \sqrt{8}}{2}$$

$$\tan \alpha = 3 \pm \sqrt{2}$$

$\therefore$ Hits target if

(1)

$$3 - \sqrt{2} \leq \tan \alpha \leq 3 + \sqrt{2}$$